Suggested solutions and discussion notes

Problem 1(a). Clean-image prediction. Rearranging the forward equation gives $\mathbf{x}_0 = (\mathbf{z}_t - b_t\boldsymbol{\epsilon})/a_t$. Replacing the unknown noise by the network prediction gives

$$\boxed{\hat{\mathbf{x}}_\theta(\mathbf{z}_t, t) = \frac{\mathbf{z}_t - b_t\boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t)}{a_t}.}$$

This formula converts any noise prediction into a clean-image prediction, even if the noise prediction has errors.

Problem 1(b). Conditional-mean prediction. Keep $\mathbf{z}_t$ and t fixed and choose a predicted image $\mathbf{u}$. The loss averages the squared error over possible clean images given $\mathbf{z}_t$ and t. Write $\mathbf{m} = \mathbb{E}[\mathbf{x}_0 \mid \mathbf{z}_t, t]$. Expanding gives

$$\begin{aligned} R(\mathbf{u}) &= \mathbb{E}[(\mathbf{x}_0 - \mathbf{u})^\top(\mathbf{x}_0 - \mathbf{u}) \mid \mathbf{z}_t, t] \\ &= \mathbb{E}[\|\mathbf{x}_0\|^2 \mid \mathbf{z}_t, t] - 2\mathbf{u}^\top\mathbf{m} + \mathbf{u}^\top\mathbf{u}. \end{aligned}$$

The first term does not depend on the chosen output. Differentiate the remaining terms:

$$\nabla_{\mathbf{u}} R(\mathbf{u}) = -2\mathbf{m} + 2\mathbf{u}.$$

Setting this to zero gives

$$\boxed{\mathbf{u} = \mathbf{x}^*(\mathbf{z}_t, t) = \mathbb{E}[\mathbf{x}_0 \mid \mathbf{z}_t, t].}$$

The Hessian is $2\mathbf{I}$, so this is the unique minimum, assuming finite second moments. A trained network may only approximate this prediction.

Problem 1(c). Predictions at different noise levels. At $t = 100$, the prediction has a distinct outline and resembles the reference zero. At $t = 700$, its strokes are fainter and less distinct, and the shape differs from the reference.

Squared error is minimised by a conditional *average*, as derived in Problem 1(b). At high noise, several clean images may have high probability given the noisy image. Their strokes need not line up, so averaging their pixel values can produce a faint or blurry image. The loss rewards this average even when it does not resemble any one likely clean image. The prediction is not a sample from the conditional distribution.

The figure uses converted predictions from a trained noise network. For an ideal squared-error noise predictor, $\boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t) = \mathbb{E}[\boldsymbol{\epsilon} \mid \mathbf{z}_t, t]$, and taking conditional expectations of the forward equation gives

$$\frac{\mathbf{z}_t - b_t\mathbb{E}[\boldsymbol{\epsilon} \mid \mathbf{z}_t, t]}{a_t} = \mathbb{E}[\mathbf{x}_0 \mid \mathbf{z}_t, t].$$

This converted prediction is also a conditional mean. The displayed predictions are approximate: model errors can also change the digit shape or introduce artefacts.

Problem 2(a). Conditional score. For fixed $\mathbf{x}_0$ and t, the Gaussian log density is

$$\log q(\mathbf{z}_t \mid \mathbf{x}_0) = -\frac{D}{2}\log(2\pi b_t^2) - \frac{1}{2b_t^2}(\mathbf{z}_t - a_t\mathbf{x}_0)^\top(\mathbf{z}_t - a_t\mathbf{x}_0).$$

Here D is the number of pixel values in the image. The first term is constant with respect to $\mathbf{z}_t$. Differentiating the quadratic and substituting the forward equation gives

$$\begin{aligned}\mathbf{h}_t(\mathbf{z}_t, \mathbf{x}_0) &= -\frac{1}{2b_t^2}\,2(\mathbf{z}_t - a_t\mathbf{x}_0) \\ &= -\frac{\mathbf{z}_t - a_t\mathbf{x}_0}{b_t^2} = \boxed{-\frac{\boldsymbol{\epsilon}}{b_t}.}\end{aligned}$$

The vector points towards $a_t\mathbf{x}_0$, the mean of the Gaussian $q(\mathbf{z}_t \mid \mathbf{x}_0)$, and is zero at that mean.

Problem 2(b). Marginal-score prediction. The loss is $\mathbb{E}[\|\mathbf{s}_\theta(\mathbf{z}_t, t) - \mathbf{h}_t(\mathbf{z}_t, \mathbf{x}_0)\|^2]$. By Problem 1(b), the prediction that minimises expected squared error at a fixed input is the conditional mean of the target:

$$\begin{aligned}\mathbf{s}^{\text{opt}}(\mathbf{z}_t, t) &= \mathbb{E}[\mathbf{h}_t(\mathbf{z}_t, \mathbf{x}_0) \mid \mathbf{z}_t, t] \\ &= \mathbb{E}[-\boldsymbol{\epsilon}/b_t \mid \mathbf{z}_t, t] \\ &= -\frac{\mathbb{E}[\boldsymbol{\epsilon} \mid \mathbf{z}_t, t]}{b_t} = \boxed{\nabla_{\mathbf{z}_t}\log q_t(\mathbf{z}_t).}\end{aligned}$$

The last equality uses the supplied identity. The training target $\mathbf{h}_t(\mathbf{z}_t, \mathbf{x}_0)$ is the conditional score for one clean image. Its conditional mean at a fixed noisy input is the marginal score $\mathbf{s}_t^*(\mathbf{z}_t)$. The network $\mathbf{s}_\theta(\mathbf{z}_t, t)$ approximates this mean.

The score is a gradient with respect to $\mathbf{z}_t$ and describes local changes in $\log q_t(\mathbf{z}_t)$.

Problem 2(c). Reverse mean in score form. Substitute $\boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t) = -b_t\mathbf{s}_\theta(\mathbf{z}_t, t)$ into the lecture's reverse mean:

$$\begin{aligned}\boldsymbol{\mu}_\theta(\mathbf{z}_t, t) &= \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{z}_t - \frac{1-\alpha_t}{b_t}(-b_t\mathbf{s}_\theta(\mathbf{z}_t, t))\right) \\ &= \boxed{\frac{1}{\sqrt{\alpha_t}}\big(\mathbf{z}_t + (1-\alpha_t)\mathbf{s}_\theta(\mathbf{z}_t, t)\big).}\end{aligned}$$

We can calculate the score estimate from the existing noise network without further training. The conversion keeps the mean unchanged. If the reverse variance is also unchanged, the sampling distribution stays the same. Using the same random noise gives the same sequence of images, apart from rounding errors. Generation does not need the original clean image or the noise added during training.

Problem 3(a). Squared-error identities. Substitute the forward equation into the clean-image conversion:

$$\begin{aligned}\hat{\mathbf{x}}_\theta(\mathbf{z}_t, t) - \mathbf{x}_0 &= \frac{a_t\mathbf{x}_0 + b_t\boldsymbol{\epsilon} - b_t\boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t)}{a_t} - \mathbf{x}_0 \\ &= -\frac{b_t}{a_t}(\boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t) - \boldsymbol{\epsilon}).\end{aligned}$$

The score conversion similarly gives

$$\mathbf{s}_\theta(\mathbf{z}_t, t) - \mathbf{h}_t(\mathbf{z}_t, \mathbf{x}_0) = -\frac{\boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t) - \boldsymbol{\epsilon}}{b_t}.$$

Squaring the norms gives $L_x = (b_t^2/a_t^2)L_\epsilon$ and $L_s = L_\epsilon/b_t^2$. Rearranging,

$$\boxed{L_\epsilon = \frac{a_t^2}{b_t^2}L_x = b_t^2 L_s.}$$

Problem 3(b). Equivalent loss weights. For the same training examples, the noise loss is

$$J_\epsilon(\theta) = \mathbb{E}\left[\frac{a_t^2}{b_t^2}L_x\right] = \mathbb{E}[b_t^2 L_s].$$

Multiply L_x by a_t^2/b_t^2 and L_s by b_t^2. Without these weights, the losses differ. Using L_x alone is equivalent to multiplying L_ϵ by b_t^2/a_t^2, giving more weight to high noise levels. Using L_s alone multiplies L_ϵ by $1/b_t^2$, giving more weight to low noise levels.

At a fixed noise level, a positive loss weight leaves the conditional-mean prediction unchanged. When one network learns to predict at all noise levels, the loss weights change which levels contribute more to training.

Problem 3(c). Output comparison.

Output	Use	Limitation
Noise	Its target is standard normal at every level, with variance 1 for each pixel.	Converting a noise error into an image error multiplies it by b_t/a_t, which is large at high noise.
Clean image	Easy to view; errors or constraints on pixel values can be expressed directly.	At high noise, squared-error predictions can average different plausible images.
Score	Can be used directly in the sampling formula in Problem 2(c).	Training targets $-\boldsymbol{\epsilon}/b_t$ have variance $1/b_t^2$ for each pixel, which grows at low noise levels.

No output is always more accurate. One noise network can provide all three predictions. Separately trained networks can differ because of their loss weights, network size, training method and sensitivity to numerical errors.

Optional solutions

Optional A. Marginal-score identity. Bayes' rule gives

$$q(\mathbf{x}_0 \mid \mathbf{z}_t, t) = \frac{p_{\text{data}}(\mathbf{x}_0) q(\mathbf{z}_t \mid \mathbf{x}_0)}{q_t(\mathbf{z}_t)}.$$

Moving the gradient inside the integral gives

$$\begin{aligned}
\nabla_{\mathbf{z}_t} \log q_t(\mathbf{z}_t) &= \frac{1}{q_t(\mathbf{z}_t)} \nabla_{\mathbf{z}_t} \int p_{\text{data}}(\mathbf{x}_0) q(\mathbf{z}_t \mid \mathbf{x}_0)\, d\mathbf{x}_0 \\
&= \frac{1}{q_t(\mathbf{z}_t)} \int p_{\text{data}}(\mathbf{x}_0) \nabla_{\mathbf{z}_t} q(\mathbf{z}_t \mid \mathbf{x}_0)\, d\mathbf{x}_0 \\
&= \int \frac{p_{\text{data}}(\mathbf{x}_0) q(\mathbf{z}_t \mid \mathbf{x}_0)}{q_t(\mathbf{z}_t)} \nabla_{\mathbf{z}_t} \log q(\mathbf{z}_t \mid \mathbf{x}_0)\, d\mathbf{x}_0 \\
&= \int q(\mathbf{x}_0 \mid \mathbf{z}_t, t) \mathbf{h}_t(\mathbf{z}_t, \mathbf{x}_0)\, d\mathbf{x}_0 \\
&= \boxed{\mathbb{E}[\mathbf{h}_t(\mathbf{z}_t, \mathbf{x}_0) \mid \mathbf{z}_t, t].}
\end{aligned}$$

For a distribution over training images, replace the integral by a sum. Each image is weighted by its conditional probability $q(\mathbf{x}_0 \mid \mathbf{z}_t, t)$.

Optional B(a). Reverse mean in noise form. Substitute $\hat{\mathbf{x}}_\theta(\mathbf{z}_t, t) = (\mathbf{z}_t - b_t \boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t))/a_t$ and collect terms:

$$\boldsymbol{\mu}_\theta(\mathbf{z}_t, t) = \left(\frac{c_t}{a_t} + d_t \right) \mathbf{z}_t - \frac{c_t b_t}{a_t} \boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t).$$

Using $a_t = \sqrt{\alpha_t}\sqrt{\bar{\alpha}_{t-1}}$ and $b_t^2 = 1 - \bar{\alpha}_t$,

$$\begin{aligned}
\frac{c_t}{a_t} + d_t &= \frac{1 - \alpha_t + \alpha_t(1 - \bar{\alpha}_{t-1})}{\sqrt{\alpha_t}(1 - \bar{\alpha}_t)} = \frac{1}{\sqrt{\alpha_t}}, \\
\frac{c_t b_t}{a_t} &= \frac{(1 - \alpha_t) b_t}{\sqrt{\alpha_t}(1 - \bar{\alpha}_t)} = \frac{1 - \alpha_t}{\sqrt{\alpha_t}\, b_t}.
\end{aligned}$$

Substituting these two coefficients gives the noise-prediction reverse mean in Problem 2(c).

Optional B(b). Gaussian KL weights. The $d_t \mathbf{z}_t$ terms cancel:

$$\tilde{\boldsymbol{\mu}}_t(\mathbf{z}_t, \mathbf{x}_0) - \boldsymbol{\mu}_\theta(\mathbf{z}_t, t) = c_t(\mathbf{x}_0 - \hat{\mathbf{x}}_\theta(\mathbf{z}_t, t)).$$

Hence

$$\begin{aligned}
D_{\text{KL}} - C_t &= \frac{c_t^2}{2\sigma_t^2} L_x \\
&= \frac{c_t^2 b_t^2}{2\sigma_t^2 a_t^2} L_\epsilon = \frac{(1 - \alpha_t)^2}{2\sigma_t^2 \alpha_t b_t^2} L_\epsilon \\
&= w_t L_\epsilon = \boxed{w_t b_t^2 L_s.}
\end{aligned}$$

These equalities hold for converted predictions at $t \geq 2$. The final reconstruction term uses the image likelihood, as in Lecture 11.